

Collaborative AI Care: Proposing an FDA-Owned Benchmark Dataset for LLM Mental Health Referral Evaluation

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Introduction.....	2
Related Works.....	3
LLMs for Mental Healthcare.....	3
Dataset-Oriented Research.....	4
Proposed Method.....	4
Dataset.....	4
Data Collection.....	4
Data Annotation.....	5
Data Storage & Security.....	5
Metrics for referral decision.....	5
Mental Health Pattern Definition.....	6
Submission-Based Benchmarking System.....	7

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LLM Safety Approval.....	7
Reference.....	8

Introduction

Large language models (LLMs) are increasingly deployed for emotional support and mental healthcare purposes, offering scalable solutions to address growing mental health demands. Tools like ChatGPT and specialized models such as Mental-LLM [7] have demonstrated potential in detecting conditions like depression and providing empathetic responses, with studies reporting significant reductions in distress ($g = 0.7$) through conversational agents [1]. These advancements leverage LLMs' ability to process natural language, making them valuable for triaging, monitoring, and supporting users in real-time, particularly in underserved populations where access to care is limited [4]. However, their widespread adoption raises critical safety and efficacy concerns, necessitating robust evaluation frameworks to ensure responsible use.

For major mental health services—such as managing severe depression, suicidality, or psychosis—human consultants remain indispensable due to LLMs' limitations in clinical reasoning, ethical judgment, and real-time crisis management. Research highlights LLMs' inconsistent referral behavior, such as delayed escalation in suicidal scenarios [6], and their inability to navigate nuanced ethical risks like bias or privacy breaches [5]. Human expertise provides the contextual understanding and accountability that LLMs lack, making LLM-human collaboration the optimal direction for mental healthcare delivery. This hybrid approach leverages LLMs' scalability for initial engagement while ensuring human oversight for complex or high-stakes cases, aligning with calls for responsible AI integration [2].

A key barrier to achieving this collaboration is the absence of a standardized, authoritative dataset to evaluate LLMs' referral capabilities—specifically, their ability to determine when to refer users to human consultants during a conversation and to do so promptly. We propose an FDA-owned black box benchmark dataset, constructed from raw data collected from consenting patients, to address this gap. This dataset will include dialogues and posts annotated for referral triggers (e.g., suicidal ideation requiring immediate escalation) and timing (e.g., referral at the point of crisis mention), enabling evaluation of whether an LLM can identify and act on critical moments, such as a user stating, “I want to end it.” By providing a controlled, regulatory-backed benchmark, companies can upload their LLMs for assessment, receiving a report on performance against specific thresholds for mental health categories and patterns (e.g.,

accuracy in suicidality referral). This referral dataset is both necessary and foundational to constructing a safe, effective LLM-human consultation framework, offering multiple benefits: (1) it ensures LLM-human collaboration achieves high clinical safety by validating timely and accurate referrals, reducing risks like missed crises; (2) it motivates LLM builders to enhance their models' referral precision and responsiveness to meet FDA standards, fostering innovation; and (3) it boosts LLM-based applications by increasing user trust—applications using LLMs that pass one or more benchmarks will reassure users of their reliability, encouraging adoption in mental healthcare settings.

Related Works

The growing use of large language models (LLMs) in mental healthcare has prompted extensive research into their capabilities and safety challenges, particularly regarding referral to human consultants. This section reviews prior work in two key areas: LLM applications in mental healthcare and dataset-oriented efforts supporting these advancements. These studies collectively highlight the need for a standardized, regulatory-backed benchmark to evaluate referral accuracy and timeliness, as proposed in our FDA-owned dataset.

LLMs for Mental Healthcare

LLM applications in mental health range from detection to conversational support, yet their referral mechanisms remain inconsistent. [7] Xu et al.'s Mental-LLM, fine-tuned on datasets like CSSRS-Suicide, achieved 87% accuracy in suicide risk prediction but lacked systematic escalation protocols, relying on ad hoc human oversight. Similarly, [8] Yang et al.'s ChatCounselor, leveraging Psych8k therapy transcripts, improved empathy but offered no clear referral framework. [6] Saha et al. evaluated ChatGPT, finding it escalated only at severe PHQ-9 levels (e.g., ≥ 20), with crisis resources provided in <50% of cases, underscoring timing issues. [1] Abd-Alrazaq et al.'s meta-analysis of 15 trials reported conversational agents reducing depression ($g = 0.64$) and distress ($g = 0.7$), yet safety measures like suicide alerts appeared in only 15 of 35 studies. [9] Thieme et al.'s review of generative AI in psychiatry noted high performance but risks like promoting harmful behavior, with referral often absent. [10] Milne-Ives et al. scoped AI's impact on mental healthcare tasks, finding diagnostic support but limited real-world referral validation. [4] D'Alfonso highlighted LLMs' diagnostic potential, yet their lack of explainability necessitates human intervention. [13] Denecke et al. examined online mental healthcare, identifying ethical gaps in crisis management, while [14] Vaidyam et

al. proposed “Artificial Wisdom” for compassionate AI, lacking practical referral triggers. [15] Kabir et al. emphasized AI’s role in student mental health, calling for safety standards. These efforts reveal LLMs’ scalability but inconsistent referral behavior, necessitating a standardized evaluation framework.

Dataset-Oriented Research

Datasets have driven LLM advancements in mental health, though their design limits referral-specific insights. [3] Cohan et al.’s SMHD provides multi-condition annotations from Reddit, enabling broad analysis but lacking referral labels. CSSRS-Suicide, used by [7] Xu et al., offers suicidality annotations, yet focuses on severity rather than binary “refer or not” decisions. DAIC-WOZ, with multimodal depression data, supports detection but not conversational timing. [11] Rashkin et al.’s EmpatheticDialogues trains empathetic responses, yet omits escalation markers. [12] Losada et al.’s eRisk series targets early risk detection (e.g., depression, anorexia), but its static posts miss dynamic referral needs. [16] Gaur et al.’s Depression Reddit Dataset provides unlabeled posts for sentiment analysis, insufficient for referral training. [17] Pisani et al.’s Crisis Text Line dataset, though restricted, offers crisis dialogue insights, yet lacks public access for broad use. These datasets excel in detection and severity tasks but fall short in supporting real-time referral evaluation, highlighting the need for a purpose-built benchmark like our proposed FDA dataset, which integrates patient-contributed dialogues with referral-specific annotations.

Proposed Method

Dataset

Data Collection

We recommend the FDA establish a secure, benchmark dataset for evaluating LLM safety in mental health applications. This data can be procured by the FDA by partnering with telehealth platforms already utilizing AI for session transcription and mental health assessment materials with their patients, particularly like the GAD-7 and PHQ. Patient data collection would require explicit informed consent, with clear explanations of how the FDA will use the information, particularly from mental health assessments. It must be emphasized that this data is solely for risk reduction and benchmarking, not model training, and will be strictly protected, accessible only to FDA personnel managing the evaluation.

Data Annotation

To enable precise evaluation of the dataset, an annotation process will be developed for dialogues or posts from the patients, incorporating a binary “refer or not” label alongside the location-specific markers. Each sample — whether a single post (e.g., from CSSRS-Suicides) or a multi-turn dialogue (e.g., from DAIC-WOZ) — will be reviewed by mental health experts to assign a ‘refer’ label if referral criterion is met. (e.g., suicidality, severe symptoms), or ‘not refer’ otherwise. Additionally, annotators will tag the exact text segment triggering the referral (e.g., “I want to end it” at sentence 3). Providing temporal or positional context within the input. This annotation process enables the evaluation of LLM models by ensuring they not only trigger referrals correctly but also do so at the appropriate time.

Data Storage & Security

Use a trusted cloud storage service like Amazon Web Services (AWS), Microsoft Azure, or Box, known for handling sensitive data. Lock the data with strong encryption, both when it’s stored and when it’s moved, so only the evaluation team can unlock it with a special key. Keep this key in a separate, safe place, like a digital vault, so even the cloud provider can’t access the data.

Set up strict rules so only the evaluation team can get in, using passwords plus an extra step, like a code sent to their phone. Make sure every time someone looks at the data, it’s recorded, so you can check who did what. Split the data into separate, locked sections to limit damage if something goes wrong. Back it up in another secure spot, also encrypted, and test it now and then to make sure it works if needed. Since this is FDA-related, pick a service that follows government health rules (like HIPAA) and check it regularly to stay compliant.

Metrics for referral decision

To assess the superset’s effectiveness, we propose a multi-faceted evaluation framework:

METRICS NAME	DESCRIPTION
Precision	Percentage of referral decisions correctly identifying cases requiring human intervention (e.g., CSSRS Level 4–5), targeting ~70% to balance specificity.

Recall	Percentage of actual high-risk or referral-worthy cases correctly flagged, prioritizing >90% to minimize missed escalations, critical for safety [6].
False Positive Rate	Percentage of low-risk cases unnecessarily referred, aiming for <20% to maintain usability.
User Trust Score	Post-interaction survey metric (e.g., 1–5 scale) assessing user confidence in referral timing, inspired by JMIR Mental Health surveys (2024).
Response Time	Average latency from risk detection to referral output, targeting <10 seconds for real-time applicability.

Mental Health Pattern Definition

We propose to extract mental health categories and patterns from existing datasets. This approach leverages established resources to evaluate LLM's behaviors for referring users to human mental health consultants. Specifically, pattern extraction from existing datasets involves following actions:

1. **Sources:** to collect the mental health category and pattern information from annotations of established datasets such as CSSRS-Suicide, DAIC-WOZ and SMHD.
2. **Analysis:** to apply natural language processing (NLP) techniques to identify recurring mental health categories and patterns (e.g., mood disorders, crisis signals).
3. **Synthesis:** cross-reference extracted patterns with raw patient data to validate relevance, focusing on naturalistic expressions rather than imposed thresholds.
4. **Review:** mental health experts to review the category and pattern definition.

We propose the FDA define and require specific referral language for LLMs used in mental health assessments. Upon detecting high risk, the LLM should provide a direct and unambiguous message: 'High Risk Detected: Please seek immediate assistance from a mental health professional. [Reason, Urgency].' This standardized, critical prompt and referral to crisis services or mental health professionals could provide the necessary intervention to save lives.

Submission-Based Benchmarking System

We propose that the U.S. Food and Drug Administration (FDA) establish a benchmark challenge for evaluating large language models (LLMs) in mental health referral scenarios, akin to existing AI benchmarks in healthcare. This initiative would involve a test-only, black-box³ dataset comprising anonymized, synthetic, or securely sourced mental health-related queries and responses, ensuring no training data is released to protect patient privacy and data security. To implement this, the FDA could either develop a bespoke secure platform or leverage an existing trusted framework, such as those used in regulated clinical trials or federal data challenges, with robust encryption and access controls. This benchmark would enable the FDA to assess LLMs' ability to accurately identify mental health needs, provide appropriate referrals, and avoid harmful outputs, fostering safer integration of AI tools in mental health support while setting a regulatory standard for industry stakeholders.

The system operates through a submission-based evaluation process: (1) LLM developers train their models using their own training datasets; (2) the developers submit their trained models for benchmarking; (3) the system conducts model inference using black-box benchmarking data; (4) a report, delivered as a CSV file, is produced to demonstrate the model's performance across various mental health categories and patterns; (5) the FDA reviews the results and may grant approval for the LLM to be used in specific mental healthcare services; (6) applications utilizing the approved LLM may then display an FDA seal within their interface.

LLM Safety Approval

We believe the FDA should approve LLMs for use in specific applications after they demonstrate consistent performance above predefined thresholds. This approval would allow developers, such as those creating mental health applications, to display an FDA seal. This seal would certify that the app's LLM has met the required performance standards for its intended purpose, assuring users that they will receive appropriate and necessary support when needed.

³ **What is a “Black Box” Test Set?**

A black box test set refers to a dataset that is hidden from LLM model builders. The black box test set is kept secret by FDA or third-party approved by FDA. LLM builders cannot access its features, labels, or distribution directly. This ensures that models are evaluated on their generalization ability rather than their capacity to overfit to a known test set.

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